AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 2: Data ingestion and quality checks

Mon 9 Nov - Fri 13 Nov 2026 · Sprint 2 · Overall progress 20%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 13 Nov 2026

On track **This week in one line:** Built loaders for NASA .mat and CALCE Excel files into one validated table: 4,631 cycles from 8 cells.

1. Goal for the week

Read both raw datasets into a single cycle-level table with one schema, and validate it.

2. Day-by-day work

Day	Work done
Mon 9 Nov	Defined the unified schema: 16 columns, one row per full charge-discharge cycle.
Tue 10 Nov	NASA loader: paired every discharge with its preceding charge; interpolated EIS resistance (Re + Rct) onto discharge times.
Wed 11 Nov	CALCE loader: read 100 Arbin Excel files in date order, renumbered cycles, kept only full 1C discharges.
Thu 12 Nov	Registry with CSV cache and validation; fixed two bugs found while checking (constant resistance, rest time counted as discharge).
Fri 13 Nov	Data-quality report, dataset document, raw-SOH figure, 5 unit tests, this report.

3. Completed deliverables

- **batterylife/data/**: schema.py, nasa.py, calce.py, registry.py.
- **data/processed/cycles_raw.csv** and data_quality.json.
- **docs/dataset.md**; tests/test_data.py (5 tests).

4. Results

Dataset	Cells	Cycles	SOH range	Extra signals
NASA PCoE	B0005, B0006, B0007, B0018	636	102% to 58%	Impedance, temperature
CALCE CS2	CS2_35 to CS2_38	3,995	104% to 15%	Internal resistance
Total	8	4,631		

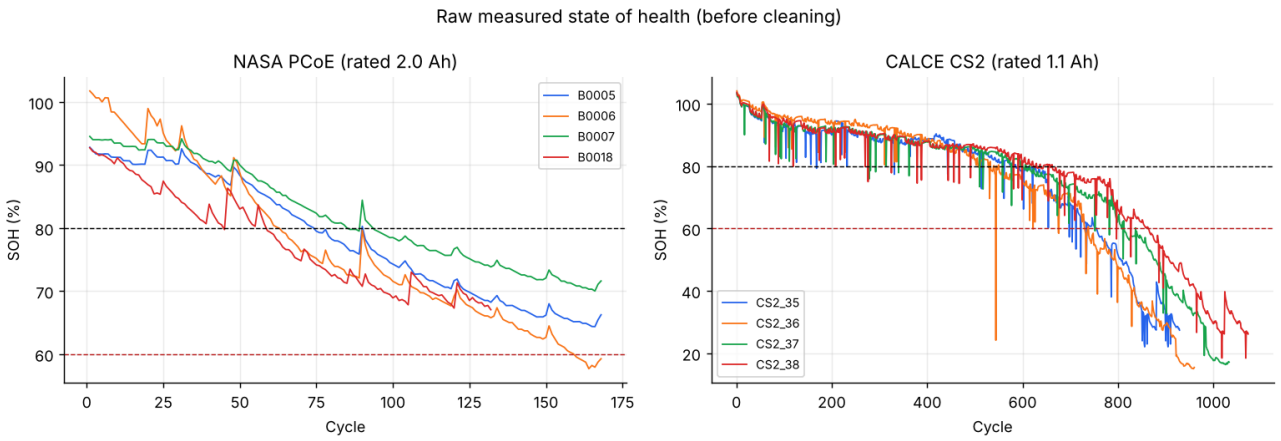


Figure 2. Measured state of health of all 8 cells before cleaning. Dashed lines: 80% and 60%.

5. Problems faced and how they were solved

Problem	Resolution
NASA resistance came out constant for every cycle.	pandas 3 stores dates in microseconds; switched to Python datetime timestamps.
NASA discharge time included rest after cut-off.	Kept only the period with load current above 1 A.
Interrupted CALCE cycles appear as sudden drops.	Planned for next week's cleaning step.

6. Plan for next week

- Remove interrupted cycles with a robust filter.
- Build health indicators from the charging curve.

Faculty remarks

Faculty signature: _____ Student signature: _____